

3. (a) Wegener's theory of continental drift was not accepted by other scientists during his lifetime because he had no explanation of how the continents moved.

We now know that the continents sit on tectonic plates which move very slowly.

State why these plates move.

[1]

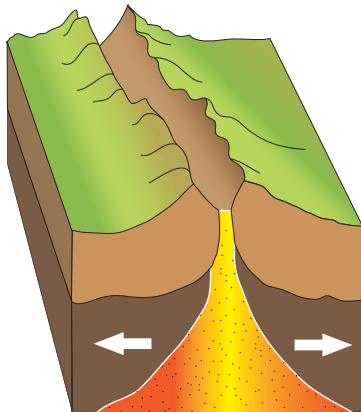
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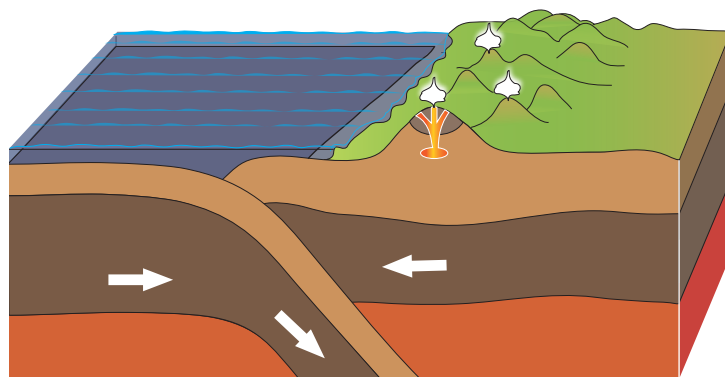
- (b) The diagrams below show two different types of plate boundary.

At a constructive plate boundary, the plates move away from each other.

At a destructive plate boundary, the plates move towards each other.



Constructive



Destructive

Describe what happens at each type of boundary.

[4]

Constructive

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Destructive

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- (c) The cities of Los Angeles and San Francisco are on opposite sides of a conservative plate boundary at a distance of 600 km apart.

They are moving closer together as the plates slide past one another at a relative speed of about 40 mm per year.

Use the formula below to calculate the amount of time before the cities are next to one another. [2]

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ m} = 1000 \text{ mm}$$

Time = years

